Section 2: Product Metrics and KPIs

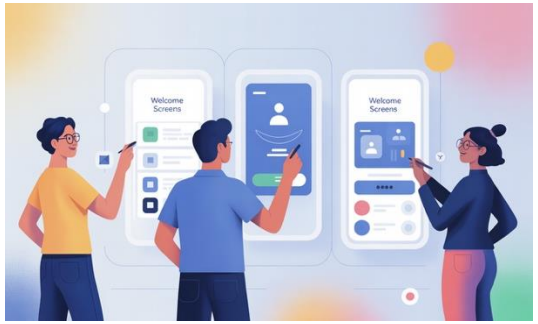
Product Analytics Course Outline

Section 1: Introduction to Product Analytics

Section 2: Product Metrics and Development

Section 3: User Acquisitions and Awareness

Section 4: User Onboarding and Activation



Section 5: User Engagement and Retention

Section 6: Product Analytics tools

Section 7: Monetization and Revenue

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Topic 1: Introduction to KPIs and Metrics

Why do we need to track metrics?

Imagine you are on a cross-country road trip and your car doesn't have a dashboard. You can’t see your current driving **speed**, **fuel** level, or other important **indicators**. Wouldn't that make you anxious?

Without these essential metrics, you might:

- Run out of gas without warning
- Exceed the speed limit
- Miss important signals like engine temperature

In much the same way, a business that doesn't track key performance metrics is operating without a "dashboard." How do you know how fast you're moving towards your goals? Are you running out of resources?

Why do we need metrics?

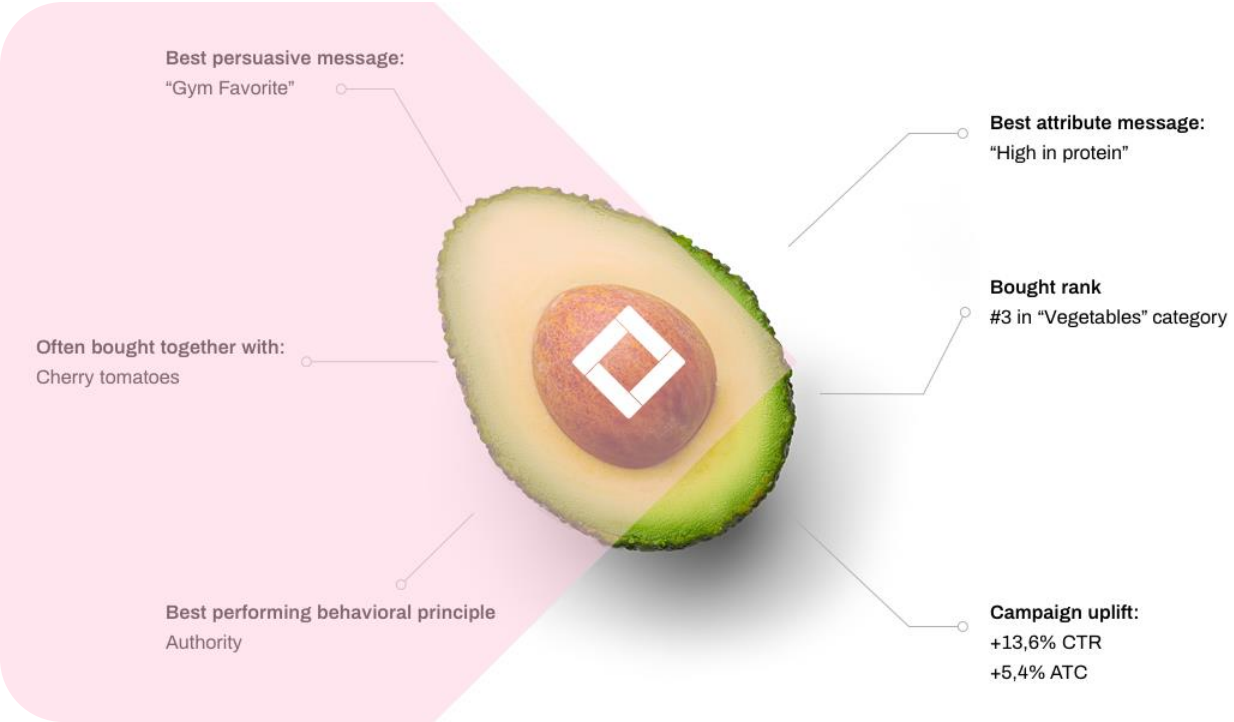
Peter Drucker said “You can’t improve what you don’t measure”

Product Analytics in Action

Product analytics involves gathering and analyzing data on user behavior to guide decisions, improve the product, and enhance user experience.

It answers three key questions:

- **What to build?** Prioritizing features based on user needs.
- **How to build it?** Designing features for maximum usability.
- **How is it performing?** Measuring success post-launch.



Product analytics enables data-driven decisions, improves user satisfaction, and helps businesses grow by focusing on what truly matters to users.

Why do we need product Analytics

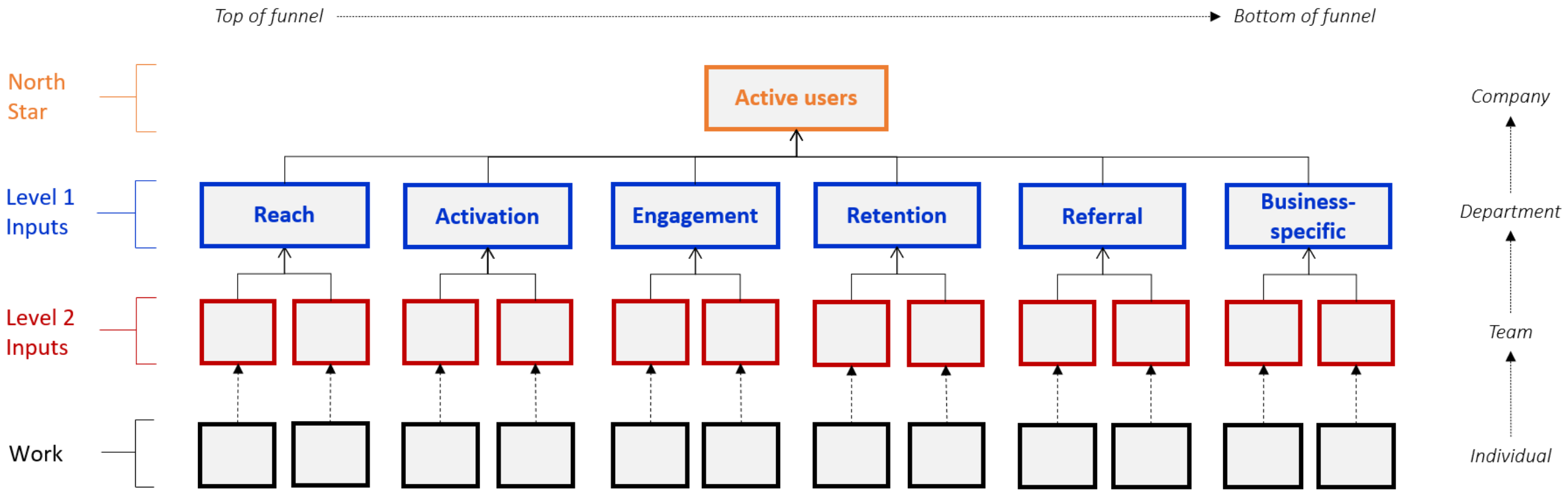
Here are some reasons why metrics are important for you to monitor as a product manager:

- To help your users discover the **aha** moment
- To help you know what features are **popular** amongst users and what features are not
- To help measure the **success** of a product feature
- To help improve customer onboarding and **retention**
- To help you determine points of customer satisfaction or dissatisfaction along the **user journey**
- To find out the number of **returning users** and infer what factors cause them to return

Topic 2: Product Metrics

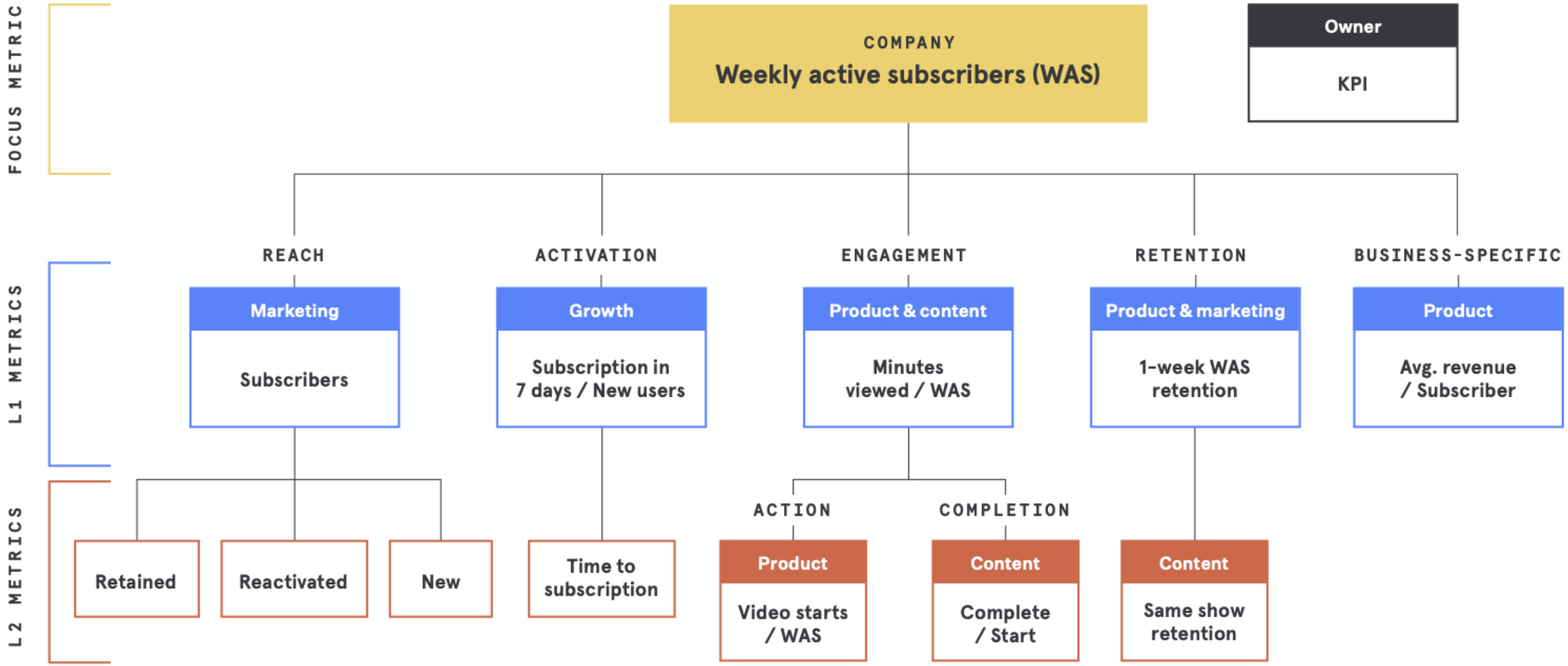
There are constellations of metrics. So not every metric can be used at the same time.

Product Metrics framework



Focus metrics (NSM) is the end Goal. So an individual achieving their goal should advance the team toward its goal, which should in turn get the department or company closer to its goals.

NorthStar Metrics : A subscription-based video streaming company



Product Metrics

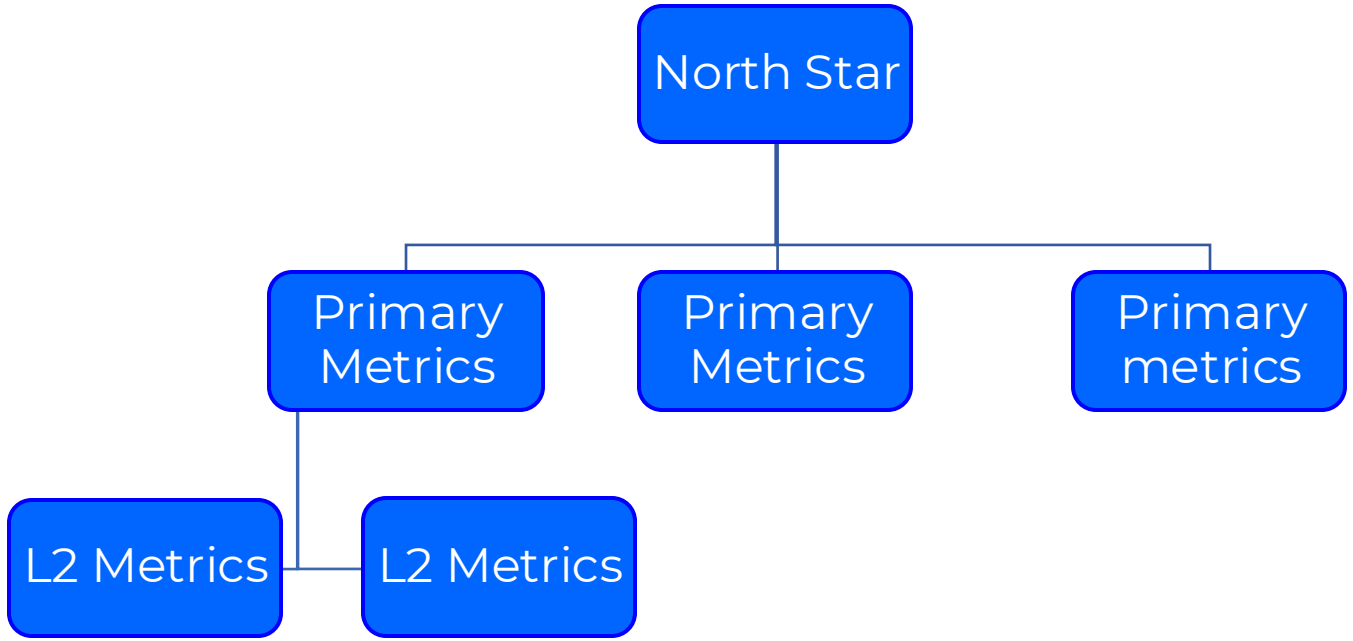
Figuring out your business goals and your product value. So here is the framework you can use to avoid tracking vanity metrics.

North Star - This reflects the prime value you give to users

Primary/Level 1 Metrics- Complement the north star/focus metric

Supporting/ Level 2 Metrics - Shows that the product is growing in a healthy direction

(Leading indicators to your north star and L2 metrics.)



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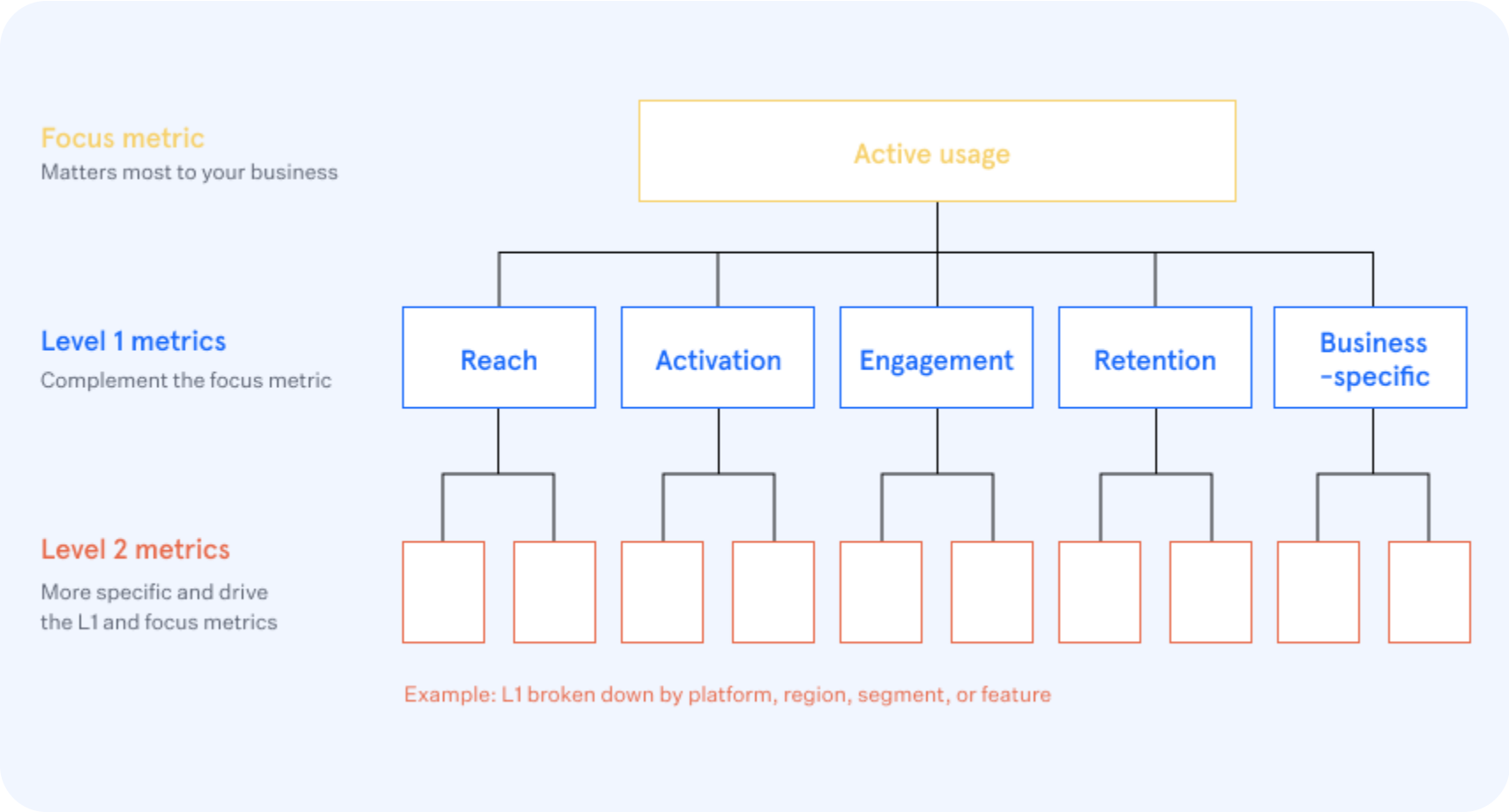
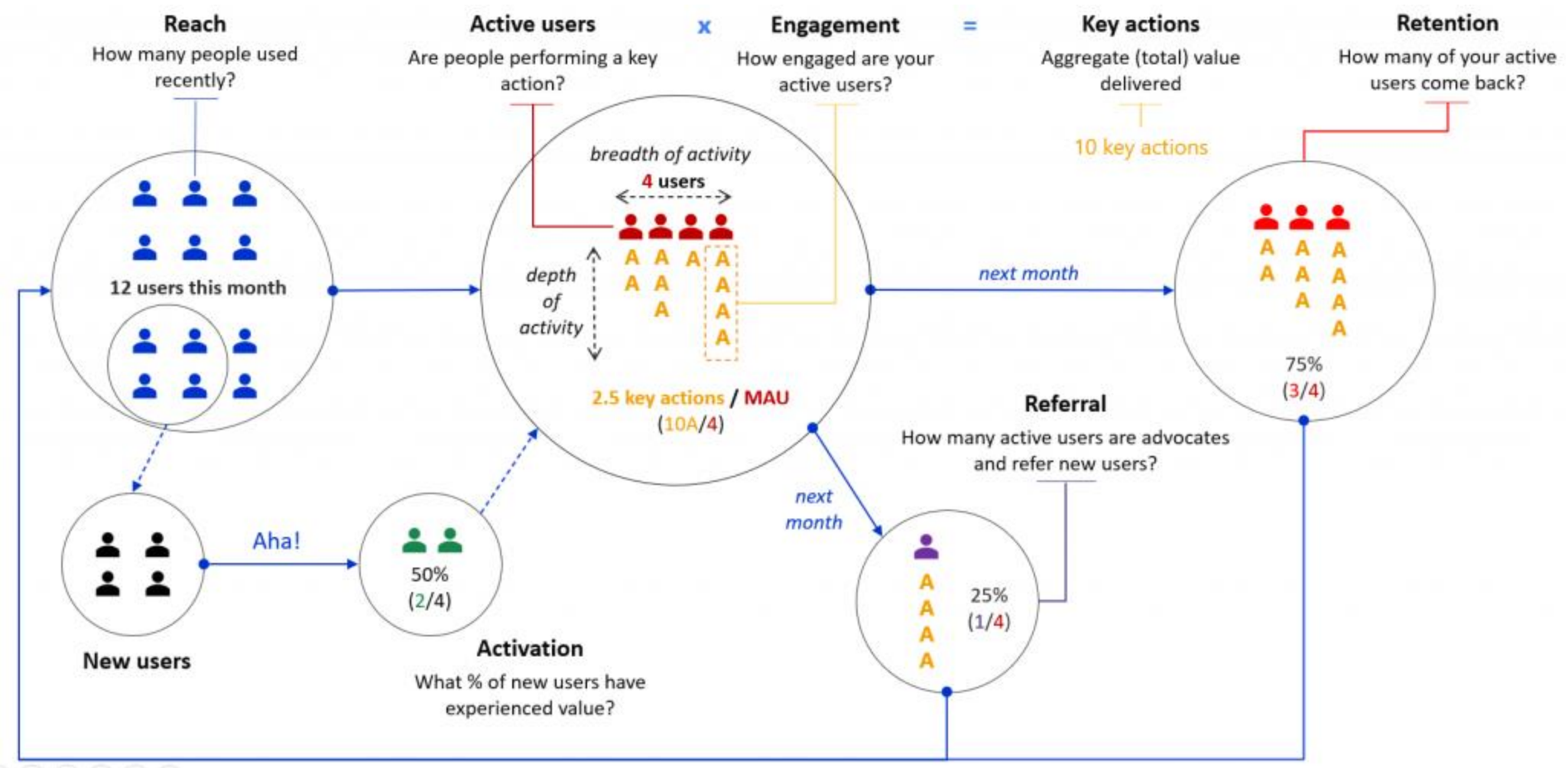
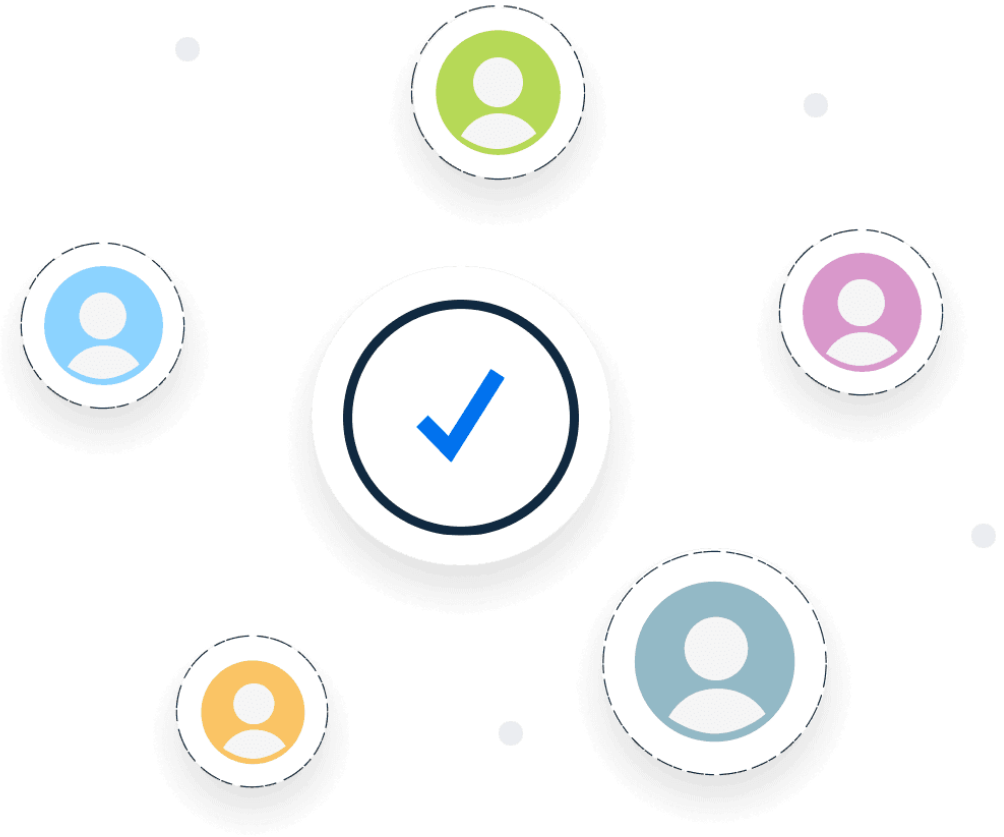


Summarizing everything



Topic 3: North Star Metrics



North Star Metrics

The most important metrics for the long-term success of your product. NSM lies in this simple statement, “If a company brings top value to its customers, then growth and success are inevitable”.

Snapchat

Daily active users

Facebook

Daily active users

Airbnb

No. of Satisfied booking

Uber

No. of Rides per week

Spotify

Time Spend everyday

Slack

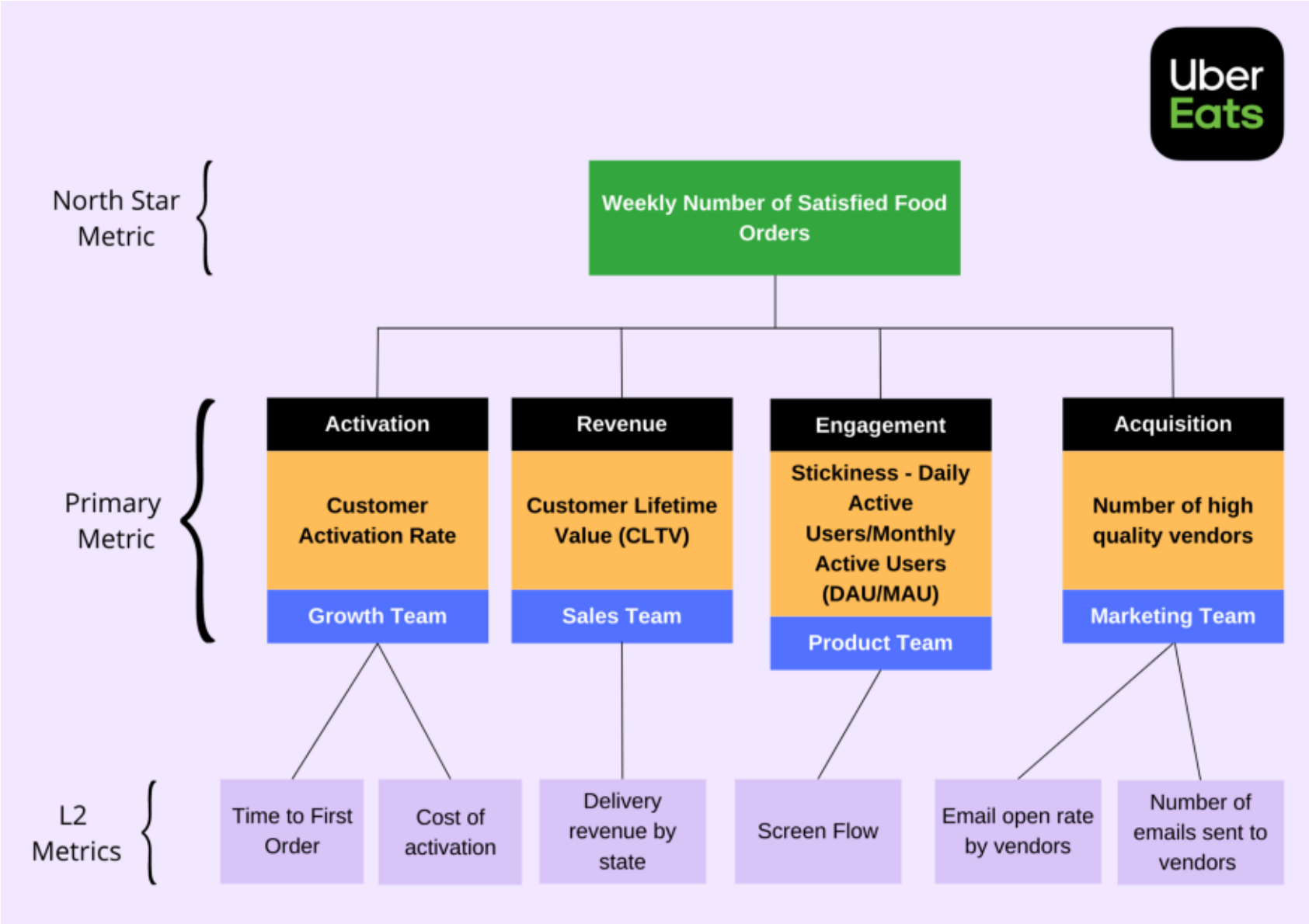
Daily Active Users

North Star Metrics

Here are some benefits of having a great North Star metric:

- **Customer Obsession:** It helps drive a mindset of customer value across the entire company.
- **Consensus:** Everyone can easily see where the company stands and can make timely re-adjustments.
- **Growth:** It offers a way to quantify the company's long-term success. Are you going to be in business for much longer or not?

North Star Metrics of Uber Eats



North Star Metrics

Primary/L1 Metrics:

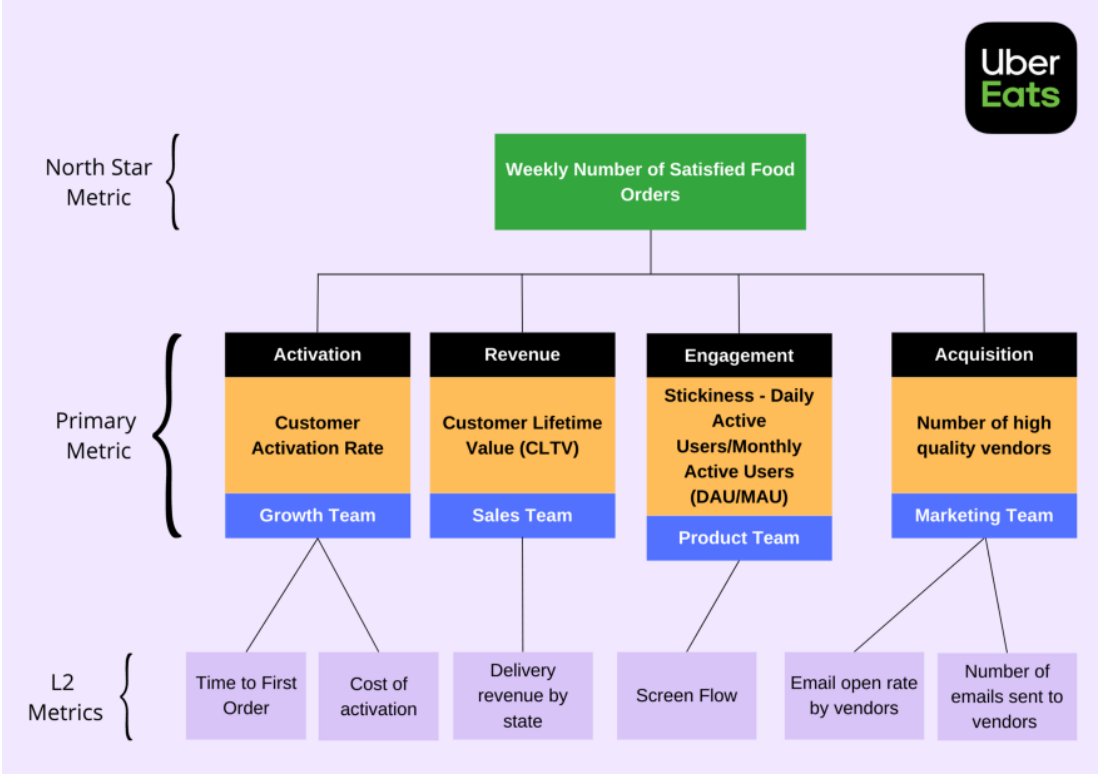
NSM is broad and focused on driving value for the users, so L1 shows the health of the focus metric.

Ex: The number of quality food vendors that sign up on the platform from their email marketing campaigns.

Supporting/L2 Metrics:

These are supporting metrics to the focus and L1 metrics. It is a way to expand on the L1 metric for each team involved.

Ex: The number of sales emails sent to food vendors, open rate, and click-through rate of the emails.



North Star Metrics

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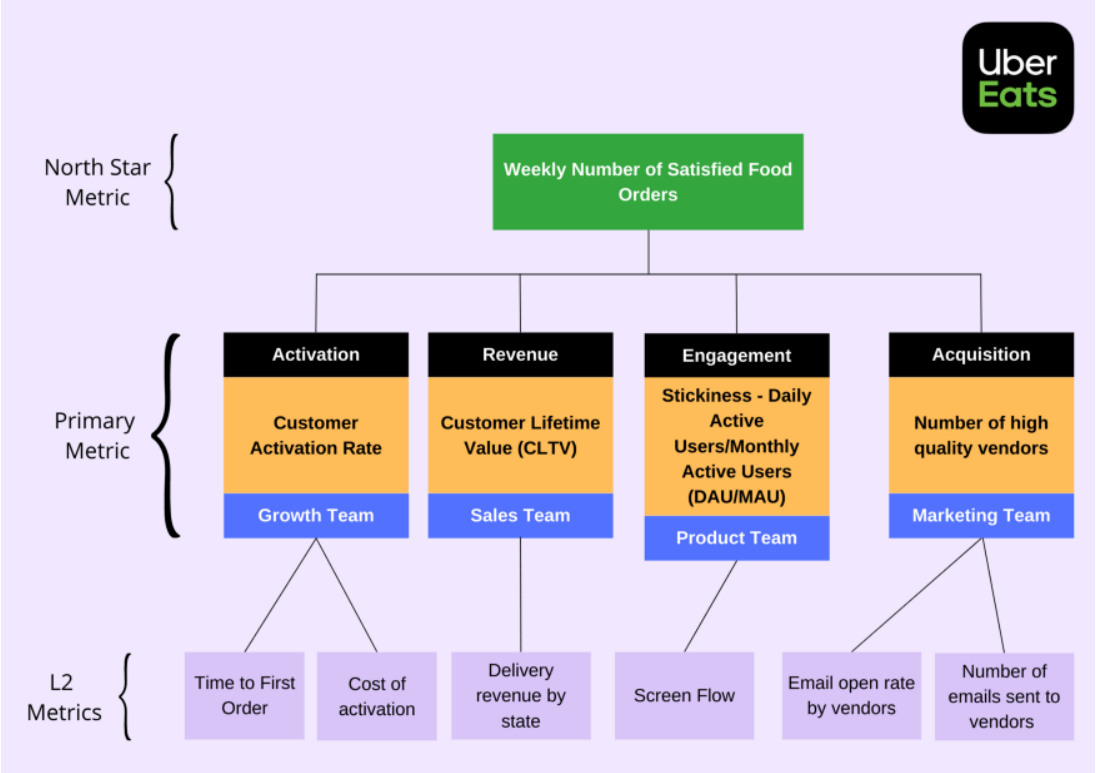
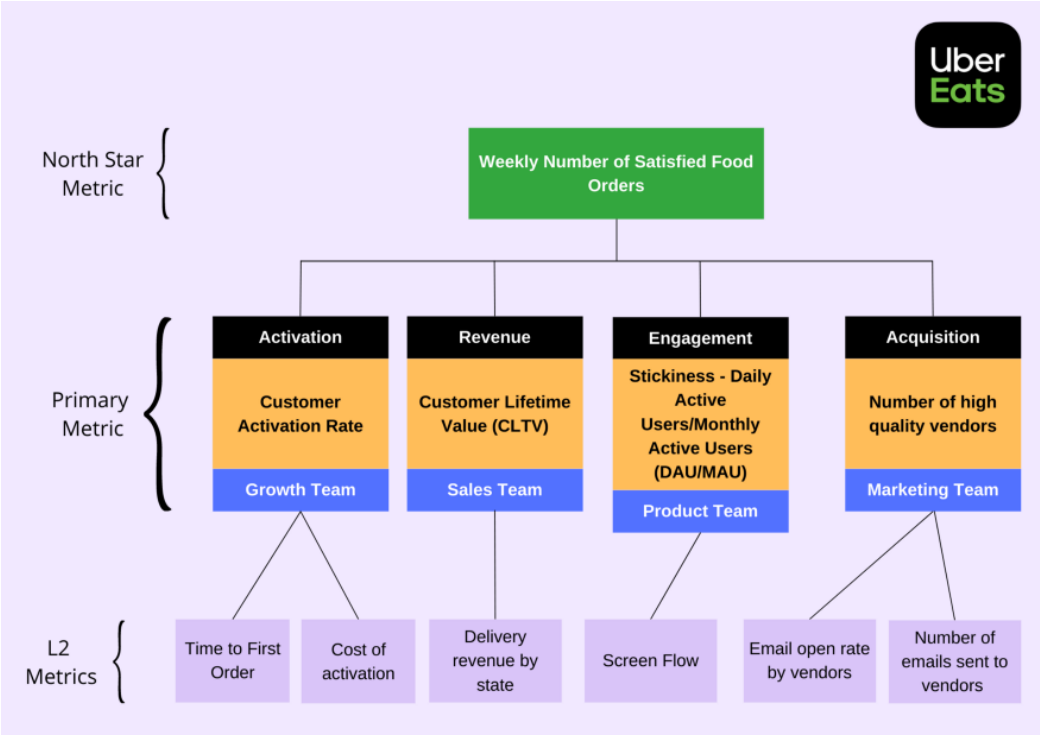


How do I choose a good North Star Metric?

North Star metrics bring focus across all the teams

1. Should be closely tied to end value proposition:



If food quality is worse than

↓

This might be the issue

↓

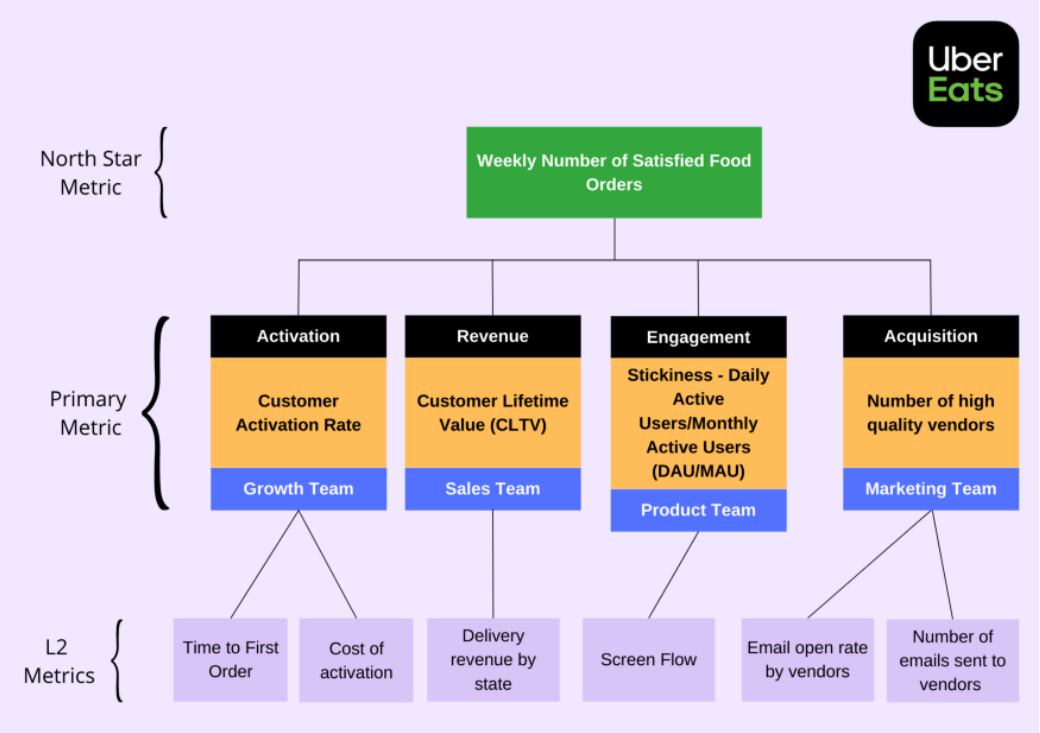
Can be solved by this

A negative review means that either the customer got wrong or the delayed delivery needs to get value from Uber Eats.

How do I choose a good North Star Metric?

2. It should be measurable:

It should be quantifiable, it should not be a feeling or state but should be actionable.



We know how many positive or negative reviews of a completed order and the contribution caused by quality and timely delivered.

How do I choose a good North Star Metric?

3. A great focus metric is time-bound: The only way to know if you are heading in the right direction is by measuring and taking account of progress in fixed periods.

Example: 10% increase in positive orders completed from the last month.

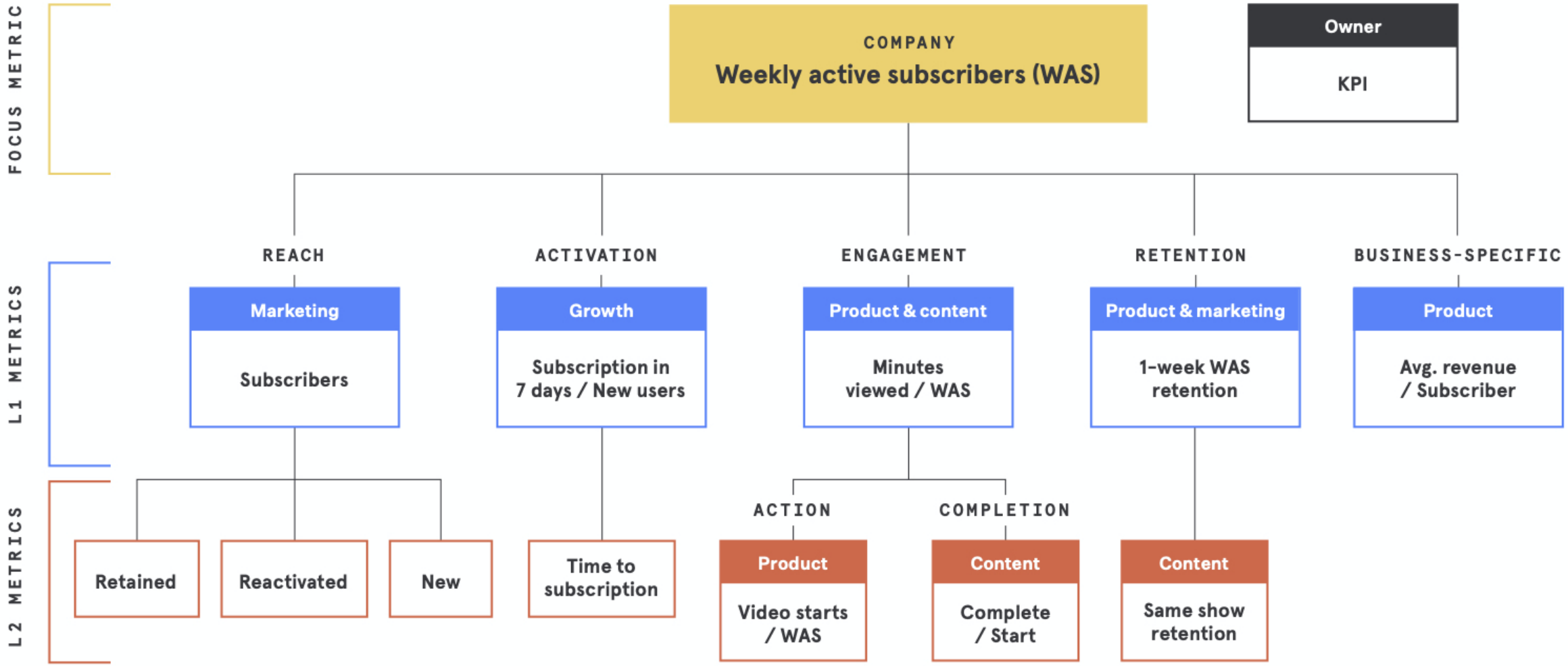
4. It can be influenced by everyone in the company:

If a change occurs on one part of the AAARRR funnel, it should ideally affect the NSM.

5. It should tie to the long-term success of the company:

A good North Star metric would point in the direction of sustainability.

NorthStar Metrics : A subscription-based video streaming company



Assignments

Mixpanel Measurement Framework

Below is a blank metrics template so you can apply the framework to your product. Print this page out and fill in the blanks. Start with a focus metric, then add the level 1 and 2 metrics.

COMPANY NAME:

PRODUCT:

Focus metric

L1 metrics

REACH	ACTIVATION	ENGAGEMENT	RETENTION	BUSINESS-SPECIFIC

L2 metrics

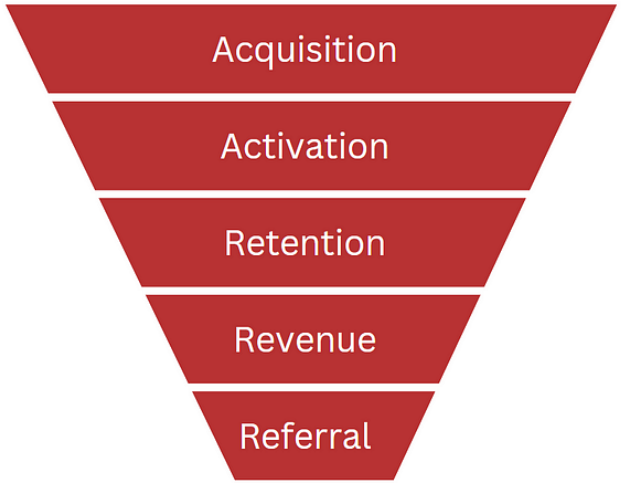
Have questions? inquiries@mixpanel.com

mixpanel

Make sure you answer these questions

1. How is it tied to the end value proposition
2. How can you measure it
3. Make it time-bound
4. Is it influenced by everyone in the company
5. Is it tied to the long-term success of the company

Topic 4: Unveiling AARRR - The Pirate Metric



Product Metrics Frameworks

The AARRR (Pirate Metrics) framework tracks the customer journey through five stages. Analyzing these stages helps product owners identify areas for improvement to boost performance.

- **Acquisition —** Tracking customer signups for a service.
- **Activation —** Getting users that have completed registration.
- **Retention —** Getting users to use the service often and behave in a way that helps the user or business.
- **Referral —** Do they like it enough to tell their friends?.
- **Revenue —** Collecting revenue from users.

AARRR Pirate Metrics framework

AARRR Pirate Metrics framework is a set of five user-behavior metrics that product-led growth businesses should be tracking

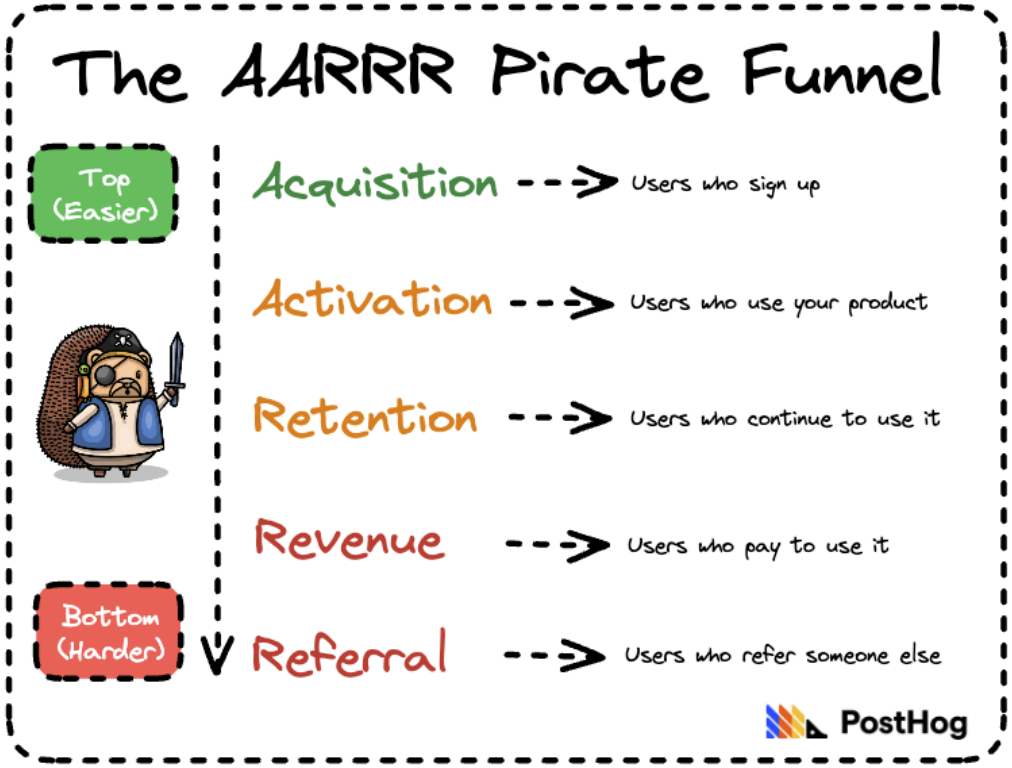
AARRR Pirate Metrics framework – Grocery Delivery

AARRR Pirate Metrics framework is a set of five user-behavior metrics that product-led growth businesses should be tracking

- **Acquisition:** New users download the app and create an account. (Sources: targeted ads, discounts for first orders)
- **Activation:** The User places their first grocery order successfully.
- **Retention:** The user places repeat orders regularly (weekly/bi-weekly).
- **Referral:** The user shares referral codes with friends, incentivizing new customer acquisition.
- **Revenue:** Sales revenue from completed orders, and potential delivery fees.

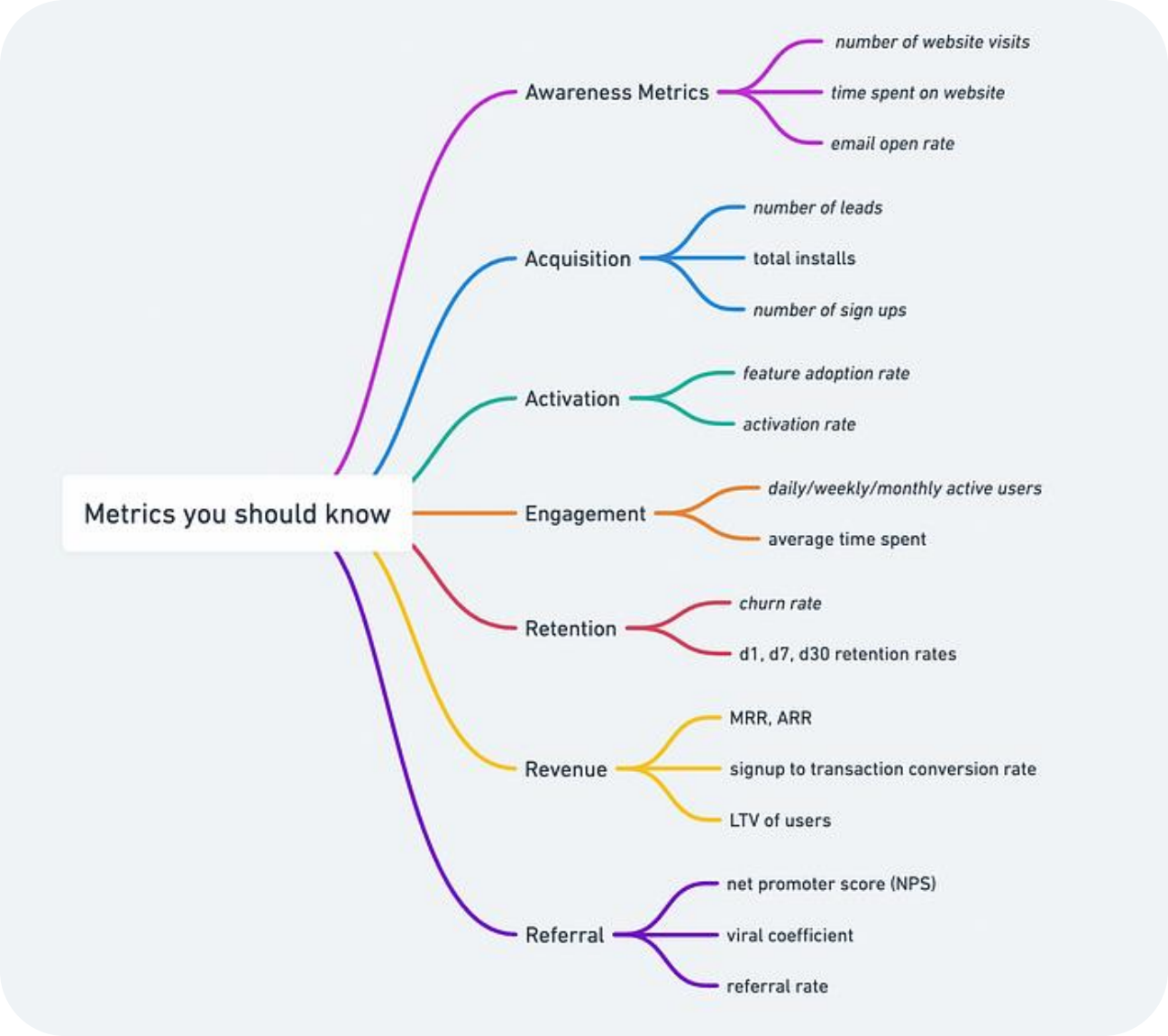
PostHog	User Status	Conversion
Acquisition	Visits website	100%
	Reaches signup page	30%
	Creates account	20%
Activation	Completes onboarding	15%
	Deploys + ingests events	10%
Retention	Active weekly usage	7%
	Invites a colleagues	4%
Revenue	Inputs card details	2%
	Generates first invoice	1.5%
Referral	Reviews product online	1%
	Refers paying customer	0.5%

AARRR Pirate Metrics framework



PostHog	User Status	Team
Acquisition	Visits website	Marketing
	Reaches signup page	Marketing / Growth
	Creates account	Growth
Activation	Completes onboarding	Growth
	Deploys + ingests events	Growth
Retention	Active weekly usage	Product
	Invites a colleagues	Growth / Product
Revenue	Inputs card details	Growth
	Generates first invoice	Growth / Customer
Referral	Reviews product online	Growth / Marketing
	Refers paying customer	Marketing

AARRR Pirate Metrics framework



Netflix to illustrate AARRR Pirate Metrics framework

Acquisition: Netflix attracts customers via social media, collaborations, and digital marketing. Key metrics include social media engagement, website traffic, and conversion rate.

Activation: New users sign up and start using features. Metrics include conversion rate, drop-off rate, and dwell time.

Retention: Netflix keeps subscribers engaged with quality content and recommendations. Metrics include churn rate, login frequency, and retention rate.

Revenue: Revenue comes from subscription fees. Metrics include customer lifetime value (CLV), expansion revenue, and monthly recurring revenue (MRR).

Referral: New users join via word-of-mouth. Metrics include the purchase rate of referred customers and viral cycle time.

Topic 5: Unveiling AARRR

Acquisition

- **First-Time Order Value:** Average basket size of first-time customers
- **Geographic Acquisition Rate:** New customer sign-ups by neighborhood/delivery zone
- **Referral Success Rate:** Percentage of existing customers who successfully refer new customers
- **First-Time User Promotion Impact:** Redemption rate of first-order discounts, Average discount value per new customer and Post-promotion retention rate
- **Download-to-Purchase Rate:** Percentage of app downloads that convert to first purchase
- **Time to First Purchase:** Average time between app installation and first order
- **Competitive Conversion:** Percentage of users switching from competitor apps

Retention & Growth Metrics

- **Customer Retention Rate:** Percentage of customers placing orders in consecutive months
- **Cart Abandonment Rate:** Percentage of users who add items but don't complete purchase
- **Premium Membership Conversion:** Percentage of customers upgrading to premium delivery service
- **Share of Wallet:** Percentage of customer's total grocery spend captured
- **Customer Acquisition Cost (CAC) Recovery Time:** Time to recover acquisition cost through margin

Business Metrics

- **Average Basket Size:** Average number of items per order
- **Category Penetration Rate:** Percentage of orders containing items from each grocery category (produce, dairy, etc.)
- **Reorder Rate:** Percentage of customers who place a second order within 30 days
- **Order Frequency:** Average number of orders per customer per month
- **Time Between Orders:** Average days between customer purchases

Customer Satisfaction Metrics

- **Post-Delivery Satisfaction:** Customer ratings after receiving their order
- **Product Freshness Rating:** Customer feedback on produce and perishable quality
- **Shopper Rating:** Customer satisfaction with personal shopper service
- **Issue Resolution Time:** Average time to resolve customer complaints
- **Net Promoter Score (NPS):** Likelihood of customers recommending the service

Financial Health Metrics

- **Average Order Value (AOV):** Average amount spent per order
- **Cost Per Delivery (CPD):** Total delivery costs divided by number of deliveries
- **Gross Margin Per Order:** Profit after cost of goods and delivery expenses
- **Fresh Product Wastage:** Value of perishable items that expire before delivery
- **Delivery Efficiency Ratio:** Number of deliveries per driver per hour

Referral & Growth

- **Referral Conversion Rate:** Percentage of referral invites that convert into first-time purchases
- **Average Referral Value (ARV):** Average order value from customers who came through referrals
- **Cost Per Referred Customer (CPRC):** Total referral reward costs divided by number of successful referrals
- **Referred Customer Retention:** Percentage of referred customers who place a second order within 30 days
- **Referral Program ROI:** Revenue generated from referred customers minus the cost of referral rewards

Dave’s original AARRR dashboard

Dave McClure’s original AARRR dashboard (Acquisition, Activation, Retention, Referral, Revenue) is often used to track user behavior and business value at each stage of the funnel.

- **Acquisition:** 100% visit the site, but only 70% stay engaged, with a low initial value ($0.01-$0.05).
- **Activation:** 30% have a "Happy 1st Visit," 5% sign up for emails, and 2% for accounts ($0.25-$3).
- **Retention:** 2%-3% engage via email or return visits, valued at $2-$5.
- **Referral:** 2% refer others and 1% lead to activations ($5-$10).
- **Revenue:** 1%-2% generate revenue, ranging from $10 to break-even at $25.

Stage	User State	Conversion %	Estimated Value
Acquisition	**Visits Site** (or landing page)	100%	$0.01
Acquisition	**Stays on Site** (views 2+ pages, stays 10+ secs, 2+ clicks)	70%	$0.05
Activation	**Happy 1st Visit** (views X pages, stays Y secs, Z clicks)	30%	$0.25
Activation	**Email Sign-up** (or anything that could lead to a repeat visit)	5%	$1.00
Activation	**Account Sign-up** (includes profile data)	2%	$3.00
Retention	**Email Open → Clickthrough**	3%	$2.00
Retention	**Repeat Visitor** (3+ visits in first 30 days)	2%	$5.00
Referral	**Refer 1+ Users Who Visit Site**	2%	$3.00
Referral	**Refer 1+ Users Who Activate**	1%	$10.00
Revenue	**User Generates Minimum Revenue**	2%	$5.00
Revenue	**User Generates Breakeven Revenue**	1%	$25.00

What is Product Analytics?

Product metrics

File Edit View Insert Format Data Tools Extensions Help

	A	B	C	D	E	F
1	**Engagement Metrics**	**Acquisition Metrics**	**Feedback Metrics**	**Usage Metrics**	**Financial metrics**	
2	Daily active users (DAU)	Number of new users	Customer satisfaction (CSAT)	Number of logins	COGS(Cost of Goods Sold)	
3	Monthly active users (MAU)	Traffic sources (e.g. organic, referral, paid)	Net Promoter Score (NPS)	Number of actions per session	Gross Profit Margin (R-COGS/R)	
4	Funnel drop-off rate	Acquisition cost	User reviews and ratings	Frequency of use	Net Profit Margin	
5	Session Time	Lead generation rate	Ticket Resolution Time	Session duration	Current Ratio	
6	Number of sessions per user	ROAS(Return on add spend) Can we divided into	Contact Rate	Number of subcategory usage	Debt-to-Equity Ratio	
7	Page views per session	sources	User feedback survey response rate	Number of transactions	Monthly recurring revenue(MRR)	
8	Click-through rate (CTR)	CPA(Cost per aquisition)		Size of basket - Commerce, No of clicks - Gaming, Content	Annual recurring revenue(ARR)	
9	Bounce rate	Retention per source	**Retention Metrics**	liked/shared/subscribed - Social		
10	Conversion rate (e.g. purchase, sign-up)	New Sign Ups	Retention rate(day 0, Day 7, Day 30 etc, depending on your platform)	Average order value (AOV)		
11	Action Metric e.g. Items added in a commerce platform, events played in a gaming platform	D0 Activation %	Churn rate (e.g. monthly, annual)	No of Subscribers		
12	Cart abandonment rate	App Open Users	Reactivation rate	Time to first Action		
13		Referrals(no of users sharing, no of referrals shared, no of activation by referrals)	Repeat purchase rate	Time to first transaction		
14	**Monetization Metrics**		Customer lifetime value (CLV)			
15	Total revenue	**Commerce Specific Metrics**	Existing user reactivation rate	**Ops Metrics**		
16	Average revenue per user (ARPU)	Time to add one item	New user reactivation rate	Inventory Days		
17		Time to build cart	Lapser reactivation rate	Working Capital		
18	Lifetime value (LTV)	Avg Selling Price(ASP)	Social Metrics No of Friends, No of likes, No of Interactions	Stock Turnover Rate		
19	Gross margin	Order Penetration by Category		Cost Per Transaction		
20	Profit margin	AOV(avg order value) by Category	**Platform Metrics**	ETA Accuracy		
21	Paying user percentage	Frequency by Category	Load times/FPS(Frame per second)	Order Accuracy		
22	Revenue by category	Discount per Order	Number of crashes	Capacity Utilisation		
23	Revenue by user segment	Brand Monetosation Revenue	Number of bugs and defects	Inventory Accuracy		
24	Avg revenue per subscriber(in case subscription is an offering)	Basket Size	Number of support tickets related to errors	Order Completion rate		
25		User Delivery Fee(UDF)	Error resolution time	Order Edit %		

Sheet1

Product metrics masterclass Video here

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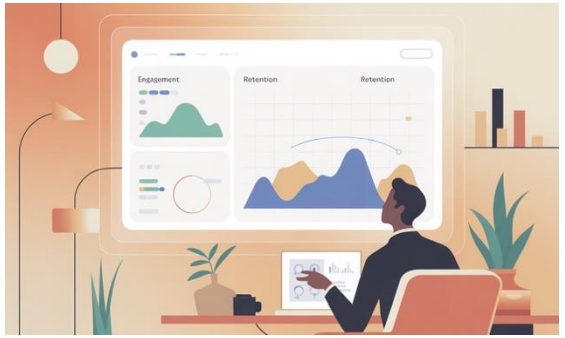

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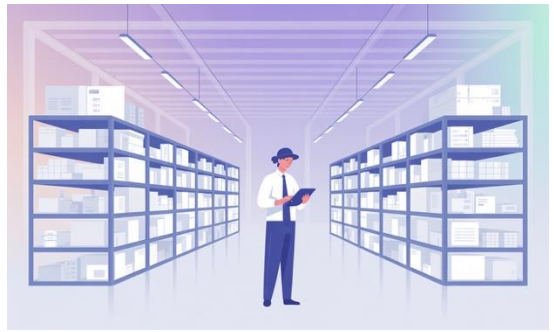

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